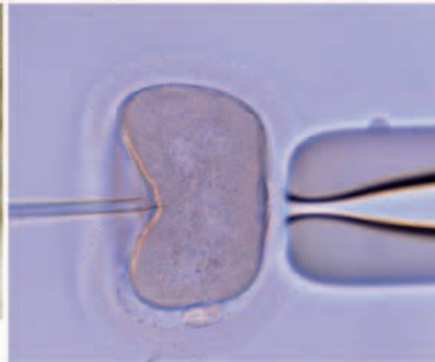




In vitro fertilization (IVF)

The first stage of the IVF artificial insemination process involves injecting sperm into an egg cell (shown above). The fertilized egg is then implanted into a woman's uterus so that a normal pregnancy process can occur.

Cholera in Bangladesh

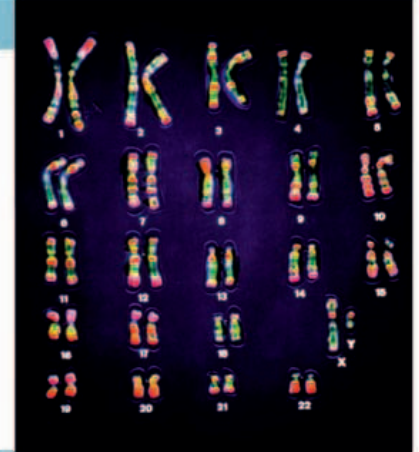
A father cradles his dying son during the 1971 cholera outbreak in Bangladesh. Lack of sanitation and access to clean drinking water led to the deaths of 6,500 people.

DISCOVERY

HUMAN GENOME

Genetics is the science of genes and heredity. Inside living cells are chromosomes. These carry genes—"instructions" that determine every aspect of an organism. Chromosomes are long molecules of deoxyribonucleic acid, or DNA. In 2000, scientists announced that they had worked out the structure of the human genome, the sequence of thousands of individual nucleic acids that make up a DNA molecule. Armed with this knowledge, doctors hope to use gene therapy to treat hereditary diseases (genetic disorders).

CHROMOSOMES OF A HUMAN MALE



Flu pandemic

In Seattle, WA, policemen wear protective masks during the outbreak of "Spanish Flu" in 1918. This was the most devastating pandemic in modern history.

fire, but the first priority is education. Public health efforts in rich countries are increasingly focused on excessive consumption of food, alcohol, and tobacco. Incidence of obesity, diabetes, heart disease, and some cancers are seriously stretching healthcare systems.

Disease pandemics

Even if the disparity between health in rich and poor nations is addressed, the world's population will still be at risk from pandemics—epidemics that occur on a global scale. Just after World War I, a pandemic of "Spanish flu" (influenza) killed up to 20 million people—more than the war itself. AIDS (Acquired Immunodeficiency Syndrome), first identified in 1981, has claimed some 25 million lives. In 2009 a flu virus crossed the species barrier from pigs, causing thousands of deaths. "Swine flu," first identified in Mexico, led to a global panic, but petered out without causing the massive death toll initially predicted.

Fears of future pandemics have also been fueled by the fact that the effects of antibiotics on disease have begun to weaken as bacteria develop immunity to drugs. "Superbugs," such as MRSA, are emerging with resistance to even the most powerful antibiotics.

3 Deaths per minute are related to inhaling cooking-fire smoke.

25 MILLION children worldwide have been orphaned by AIDS.

43 MILLION US citizens are clinically obese.

AFTER

Whether it is anticipating new pandemics or battling existing diseases, there are many challenges facing 21st-century medicine.

NEW DRUGS, NEW VACCINES

The rise of antibiotic-resistant "superbugs" means that new drugs and treatments will be needed to combat bacterial infection. There is some experimental use of bacteriophages, which are viruses that attack bacteria. Other priorities include developing vaccines against HIV/AIDS and malaria.



Malaria kills over 1 million people per year, and in some regions the malaria parasite that causes the disease is growing increasingly resistant to current drug treatments.

BACTERIOPHAGE

ADVANCES IN MEDICINE

Since the 1920s, tremendous advances have been made in medicine and surgery as scientists, researchers, and doctors seek solutions to world health problems.

